21-242 Recitation 12

Fields and Vector Spaces — Q3. - =(-1)

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

Prove that $-\vec{v} = (-1)\vec{v}$.

Course-level solution

It is enough to check that $(-1)\vec{v}$ is an additive inverse of $\vec{v}$, because additive inverses are unique (if $\vec{v}+\vec{w} = \vec{0} = \vec{v}+\vec{u}$ then $\vec{w} = \vec{0}+\vec{w} = (-\vec{v}+\vec{v})+\vec{w} = -\vec{v}+(\vec{v}+\vec{w}) = -\vec{v}+\vec{0} = -\vec{v}$, and likewise $\vec{u} = -\vec{v}$).
Now

$$\vec{v} + (-1)\vec{v} = 1\cdot\vec{v} + (-1)\vec{v} = \big(1+(-1)\big)\vec{v} = 0\vec{v} = \vec{0},$$

using Q2. Therefore $-\vec{v} = (-1)\vec{v}$.

Later-course solution (using ideas from the rest of the handout)

The coordinate proof is again $[\vec{v}] + [(-1)\vec{v}] = [\vec{v}] + (-1)[\vec{v}] = \vec{0}$ in $\mathbb{F}^n$. The axiom-level proof is the one you want on a midterm at this stage of the course.